

FORMAL LANGUAGES AND AUTOMATA THEORY

QUIZ # 3 – SOLUTION

REG#: _____

NAME: _____

COURSE CODE: CS224

INSTRUCTOR: MUHAMMAD SAJID ALI

TOTAL MARKS: 10

Question # 1: Subtraction of binary numbers is an arithmetic operation like the subtraction of decimal numbers or base 10 numbers. When you subtract several columns of binary digits, you must consider the borrowing i.e., when 1 is to be subtracted from 0, the result is 1 where 1 is borrowed from the next highest order bit or digit. Your task is to design a *Mealy Machine* that decreases a binary number by 1 with the help of the following subtraction rules and given examples. (CLO2) (5 Marks)

- Rules for binary subtraction

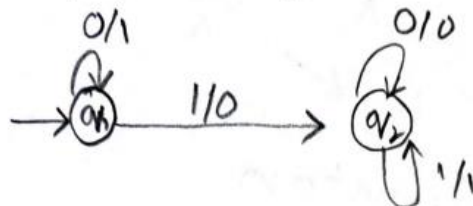
Binary Number	Subtracted Value
0 - 0	0
1 - 0	1
0 - 1	1 (Borrow 1 from the next high order digit)
1 - 1	0

- Examples of binary subtraction using the above rules.

$$\begin{array}{r}
 1010 \\
 1) \quad \underline{} \\
 1001
 \end{array}
 \qquad
 \begin{array}{r}
 1011 \\
 2) \quad \underline{} \\
 1010
 \end{array}$$

Assumptions.

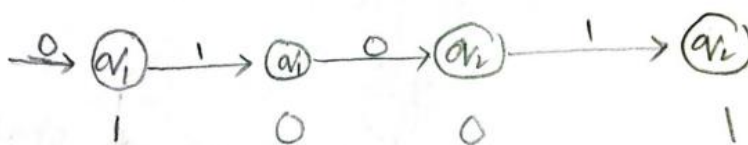
- input is read from right to left
 - Reverse output



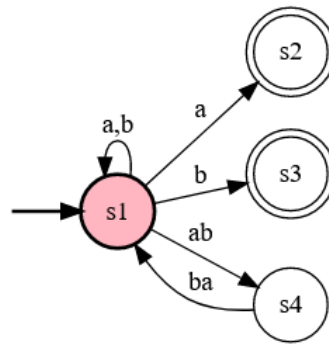
Example

$$S = 1010$$

$$\text{REVERSE}(S) = 0101$$

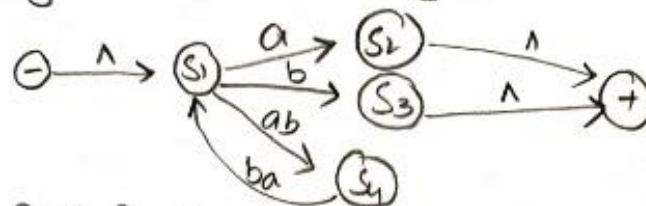


Question 2: Convert the following Transition Graph (TG) into a regular expression using Kleene's Theorem. Clearly show all of your steps. (CLO2) (5 MARKS)



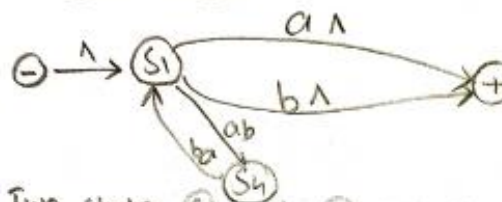
Step 1: Simplification

① only one start and ending state

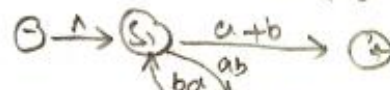


Step 2:- Build Regular Expression (R.E)

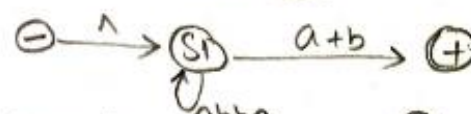
① Eliminating middle state b/w (s1) and (+) which is (s2) and (s3)



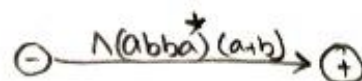
② Two states (s1) and (+) are connected with more than one edge. Change it to one edge labelling R.E to $a + b$ (ignore λ)



③ eliminate state (s4)



④ Eliminate middle state (s1)



R.E $(abba)^*(a+b)$